ADITYA UNIVERSITY

Freshman Engineering Department
Department of Mathematics
Practice Questions

AY: 2025-26 **SEM-II**

Course Name & Code: Differential Equations and Vector Calculus (2501MA02).

Unit-I: First Order & First Degree Differential Equations.

Solve the following Differential Equations (From 1 to 12):

1. $\frac{dy}{dx}+\frac{y}{x\log x}=\frac{\sin 2x}{\log x}$.
2. $\frac{dy}{dx}+2y=e^x+x, y(0)=1.$
3. $dr+(2r\cot\theta+\sin 2\theta)d\theta=0.$
4. $(1+y^2)dx=(tan^{-1}y-x)dy.$
5. $(2xy+y-\tan y)dx+(x^2-x\,tan^2y+sec^2y)dy=0.$
6. $\left(y^2e^{xy^2}+4\,x^3\right)dx+\left(2xy\,e^{xy^2}-3\,y^2\right)dy=0.$
7. $\frac{dy}{dx}+\frac{y\,cos\,x+\sin y+y}{\sin x+xcos\,y+x}=0.$
8. $(x^2y-2\,xy^2)dx-(x^3-3x^2y)dy=0.$
9. $y\,(xy\sin xy+\cos xy)dx+x(xy\sin xy-\cos xy)dy=0$
10. $\frac{dy}{dx}=\frac{x^2+y^2+1}{2\,xy}$.
11. $(xy^3+y)dx+2(x^2y^2+x+y^4)dy=0.$
12. $x^4\frac{dy}{dx}+x^3y+cosec\,(xy)=0.$
13. Find orthogonal trajectories of family of circles $x^2+y^2+2\lambda x+c=0$ where λ is a parameter.
14. Show that the following curves are self orthogonal curves.

    (i) $y^2=4a(x+a)$, a is a parameter (ii) $\frac{x^2}{a^2+\lambda}+\frac{y^2}{b^2+\lambda}=1$, λ is a parameter.
15. Find the orthogonal trajectories of the following curves:

    (i) $r=a(1-\cos\theta)$, a is a parameter (ii) $r=2a(\cos\theta+\sin\theta)$, a is a parameter.
16. Model R-L circuit, write its Differential Equation and find the current *i* at any time *t* in the circuit.
17. Find the current in the electrical circuit given by $L\frac{di}{dt}+Ri=E, i(0)=0$ where $E=6$ volts, $R=100$ ohms, $L=0.1$ henries. Also find how long will be it before the current has reached one half of its maximum value.

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